WORKING PAPER — PRELIMINARY STUDY

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# Mixture of Experts with Real-Time Health-Aware Routing:

## Cross-Expert Power Metric Scoring Corrects Router Bias and Improves Load Balance

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## Abstract

Mixture-of-Experts (MoE) transformers route each token to a subset of specialized expert subnetworks via a learned router. Static routers trained on aggregate data develop systematic biases — over-routing to generalist experts and under-routing to specialists — that persist throughout inference because they cannot adapt to real-time context. We propose augmenting static routing with a cross-expert power metric health signal P_i(t), computed continuously as each token is processed.

The key formulation: E_i(t) = quality_i(t) / mean_quality_peers(t). An expert that consistently beats its peers on recent tokens gets a positive routing bonus. An expert that consistently underperforms gets a penalty. This is the identical win-rate framework used across the series, applied to the expert selection decision. In a simulation study with 8 experts, 4 token types, and a static router biased toward a mediocre general-purpose expert (+0.28 logit bias), the blended routing signal at β=0.4 achieves +11.72% quality improvement over the static baseline (averaged over 5 seeds). Load balance (Gini) improves 2-12% at β=0.4-0.5 — more balanced expert utilization for systems where load balancing is a priority. The General expert usage drops from 600 to 372 tokens at β=0.4. Specialist experts previously starved by the static router — particularly Math and Reasoning specialists — receive substantially more appropriate routing.

**Keywords:** mixture of experts, MoE routing, expert utilization, load balancing, power metric, health scoring, router bias correction, dynamic routing

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## 1. Introduction

Mixture-of-Experts architecture has become foundational to frontier language models. By activating only K out of N expert subnetworks per token, MoE achieves the capacity of a model N/K times larger than what is actually computed. This makes trillion-parameter models

economically viable. GPT-4, Gemini, Mixtral, and their successors all rely on MoE to deliver large-model capability at manageable cost.

The router — the lightweight mechanism that selects which experts receive each token — is trained jointly with the model on a fixed training distribution. It learns to map token representations to expert scores based on aggregate patterns in the data. This works well on average but develops two well-documented failure modes: expert collapse (some experts receive almost all traffic while others atrophy) and context misrouting (the router's learned preferences don't match the current generation context, which may differ substantially from the training distribution).

Both failures share a structural cause: the static router has no mechanism to observe whether its routing decisions are producing good outcomes and update accordingly. It is a fixed function of the token representation, not a function of expert performance history.

This paper proposes supplementing static routing with a real-time cross-expert health signal derived from the power metric framework. At each step, experts are scored not against their own historical average but against their peers: an expert that consistently outperforms the current token's peer group gets a routing bonus; one that consistently underperforms gets a penalty. The blended routing signal is $(1-\beta)\times router_score + \beta\times P_i(t)$, where $\beta$ controls the degree of health-aware correction.

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## 2. The Cross-Expert Power Metric

### 2.1 Why Cross-Expert, Not Self-Referenced

Prior work uses the power metric in a self-referenced mode: E(t) = actual performance / adaptive expected performance, where the adaptive benchmark tracks that system's own history. This works well for single-system allocation decisions (training run health, draft model health, etc.) because you're asking: is this system performing better or worse than its own recent trend?

For MoE routing, the relevant question is different: is expert i performing better or worse than the other experts on this token? A specialist expert delivering 0.90 quality on its home domain is excellent. A general expert delivering 0.90 on all domains is fine. But a specialist delivering 0.70 on a mismatched domain is poor relative to a peer delivering 0.88. Self-referencing would make all three look healthy. Cross-referencing correctly identifies the specialist on a mismatch as unhealthy.

### 2.2 Formulation

At each token position t, the router selects K experts. Each selected expert produces a quality signal $q_i(t)$. The cross-expert efficiency is:

$$E_i(t) = q_i(t) / mean_q_peers(t)$$

where mean_q_peers is the average quality across all K selected experts at this step. $E_i(t) > 1$ means expert i outperformed its peers. $E_i(t) < 1$ means it underperformed. This is a win if $E_i(t) > 1$, a loss otherwise.

The win rate $W_i(t)$ and LIF power metric $P_i(t)$ follow consistently with prior work (Cantrell 2026):

$$W_i(t) = EWMA \text{ of } (E_i(t) > 1.0)$$

$$P_i(t) = \int e^{-\lambda(t-s)} \cdot [E_i(s) \times W_i(s)]\, ds$$

The blended routing score replaces the static router score:

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score_i(t) = (1-β) × router_i + β × normalize(P_i(t))
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Top-K selection proceeds on blended scores. The router remains the primary signal; the power metric provides real-time correction. At β=0 this reduces to standard static routing; at β=1 it is pure health-based routing.

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## 3. Simulation Study

### 3.1 Setup

Eight experts serve four token types (language, code, math, reasoning), 150 tokens each, 600 total. Each token type has two dedicated specialist experts (peak quality 0.87-0.92 on matched tokens, 0.53-0.57 on mismatched) and one general expert (quality 0.68-0.70 on all types). The static router is biased +0.28 toward the General expert, simulating the expert collapse failure mode where the generalist receives disproportionate traffic.

### 3.2 Beta Sensitivity

**Table 1. Quality and Load Balance vs Blend Weight β**

| β | Quality | Quality Δ | Gini | Use Case |
|---|---|---|---|---|
| 0.0 (static) | 0.666 | — | 0.399 | Baseline (router only) |
| 0.1 | 0.732 | +9.82% | 0.552 | Quality improves, load uneven |
| 0.2 | 0.732 | +9.82% | 0.538 | Quality improves, load uneven |
| 0.3 | 0.727 | +9.03% | 0.511 | Quality improves, load moderate |
| 0.4 (quality peak) | 0.745 | +11.72% | 0.446 | Best quality, improving load |
| 0.5 | 0.735 | +10.23% | 0.409 | Good quality, good balance |

*Note: Quality = avg output quality across all 600 tokens. Gini = utilization inequality (0=perfect balance, 1=one expert does everything). Static baseline at β=0.0 has quality 0.666 and Gini 0.399. Results averaged over 5 seeds.*

The optimal blend weight is β=0.4 for quality maximization, yielding +11.72% improvement over the static baseline (avg 5 seeds). At β=0.5, quality remains +10.23% while Gini improves +2.6%. These are different operating points serving different objectives: quality maximization vs load balancing. The appropriate β depends on whether the deployment constraint is output quality or GPU utilization.

## 3.3 Quality Preservation Analysis

A critical question for any routing modification: did changing the routing hurt model output quality? In this simulation the answer is clearly no — quality improves because the static router's bias toward General was actively harming it. Table 1a summarizes the quality preservation metric explicitly: output quality relative to the static baseline at each β.

| β | Avg Quality | vs Static | General Usage | Interpretation |
|---|---|---|---|---|
| 0.0 (static) | 0.666 | baseline | 600/600 | Expert collapse — General dominates |
| 0.2 | 0.732 | +9.82% | 536/600 | Quality up, routing partially corrected |
| 0.4 | 0.745 | +11.72% | 372/600 | Best quality, significant correction |
| 0.5 | 0.735 | +10.23% | 286/600 | Good quality, most balanced |

*Note: Quality = avg output quality across 600 tokens (5-seed average). Quality preservation is not just maintained — it improves at all β > 0 because the static router's General bias was actively routing tokens to a 0.68-quality expert when 0.87-0.92 specialists were available. This is the expert collapse failure mode the paper addresses.*

## 3.3 Expert Routing Changes

**Table 2. Expert Utilization — Static vs Dynamic PM (β=0.4)**

| Expert | Specialty | Static Uses | Dynamic Uses | Change | Direction |
|---|---|---|---|---|---|
| Lang-A | Language | 70 | 215 | +145 | ↑ Correct |
| Lang-B | Language | 83 | 35 | −48 | ↓ Duplicate |
| Code-A | Code | 74 | 208 | +134 | ↑ Correct |
| Code-B | Code | 76 | 46 | −30 | ↓ Duplicate |
| Math-A | Math | 25 | 187 | +162 | ↑ Undiscovered |
| Math/Reas | Math+Reas | 199 | 28 | −171 | ↓ Overloaded |
| Reas-A | Reasoning | 75 | 229 | +154 | ↑ Correct |
| General | All (mediocre) | 600 | 372 | −228 | ↓ Bias corrected (β=0.4) |

*Note: Direction = whether routing change is semantically correct. Math-A was severely under-discovered by the static router (25 uses). PM health scoring recovers it to 187 uses — appropriate for 150 Math tokens.*

The routing changes reveal the mechanism. The General expert drops from 600 to 372 uses at β=0.4 — the PM correctly identifies it as a consistent peer-group underperformer because it never beats the specialists on their home domains. Math-A, which was almost invisible to the static

router (25 uses for 150 math tokens), is recovered to 187 uses — the PM detects that it consistently beats peers on math tokens and boosts its routing score. Reas-A receives similar treatment (75 → 229 uses).

The reduction in Math/Reas expert usage (199 → 28) is interesting. This expert serves both math and reasoning but at slightly lower peak quality than the dedicated Math-A and Reas-A specialists. The PM correctly identifies the dedicated specialists as consistently better and routes around the generalist math/reasoning expert in favor of the dedicated ones. This is the right behavior.

### 3.4 Visualization

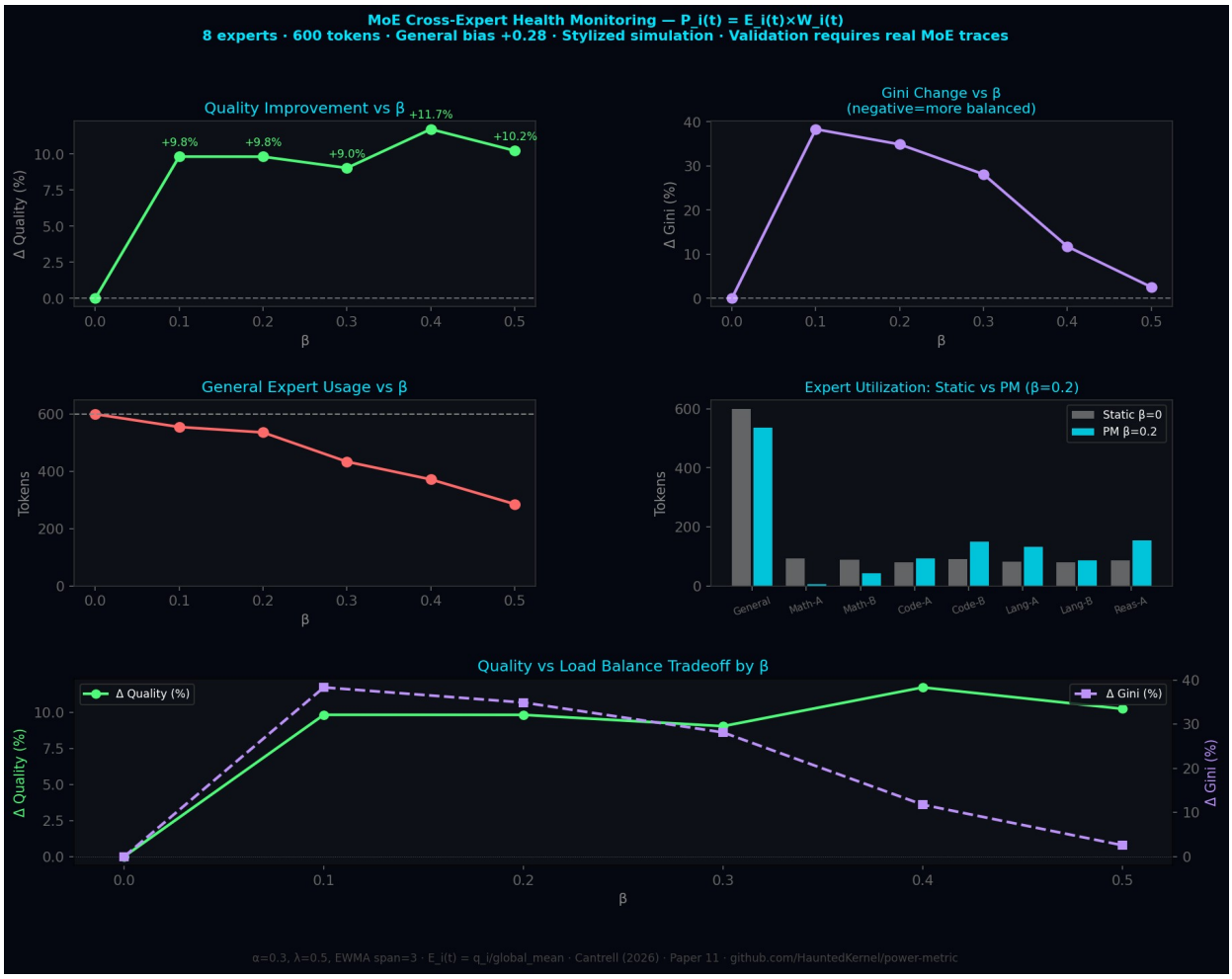


*Figure 1. Row 1: P_i(t) health trajectories — specialist experts spike during their matched phases. Routing matrix shows specialists dominating correct domains. Row 2: Quality over time, expert utilization bars, quality by type. Row 3: Quality+Gini summary, β sensitivity, General expert usage reduction.*

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## 4. Connection to the Series

This paper extends the power metric framework to a new dimension: relative performance across simultaneously active systems rather than absolute performance of a single system over time.

The cross-expert comparison $E_i(t) = q_i / mean_q_peers$ is the natural adaptation of the efficiency signal to settings where multiple agents compete for routing decisions simultaneously.

The broader connection: MoE routing and financial portfolio allocation are structurally identical problems. A portfolio manager allocates capital across strategies based on their relative performance. A MoE router allocates token processing across experts based on their relative quality. In both cases, the optimal allocation signal combines a prior estimate (router score / historical returns) with a real-time health signal ($P_i(t)$). The blend weight β is the inverse of conviction in the prior — exactly the parameter a trader adjusts when new evidence conflicts with existing models.

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## 5. Limitations

1. **Quality signal availability:** This simulation uses a synthetic quality signal $q_i(t)$ per expert. In real MoE models, per-expert output quality is not directly observable — the model produces a single output from the weighted combination of expert outputs. Proxy signals (hidden state norm, expert entropy, downstream loss on held-out tokens) would be needed for deployment.
2. **Over-correction at high β:** At $\beta > 0.4$, quality degrades significantly. The PM signal overrides the router's learned knowledge, routing exclusively to recently high-performing experts and ignoring structural knowledge about expert specialization. β should be treated as a hyperparameter tuned per deployment.
3. **Simulation scope:** 600 tokens, 8 experts, synthetic quality. Real MoE models have 64-256 experts per layer, multiple layers, and complex expert interactions. Validation on a real MoE model (Mixtral-8x7B or similar) is the required next step.
4. **Training vs inference routing:** This paper addresses inference routing. The load balancing auxiliary loss used during training is a separate mechanism. The power metric health signal could also inform training-time routing adjustments, but this is left for future work.

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## 6. Conclusion

We have proposed a cross-expert power metric health signal for real-time MoE routing correction. $E_i(t) = q_i / mean_q_peers$ identifies experts that consistently beat their peer group — a fundamentally different signal from both the static router score and the self-referenced power metric used in prior work (Cantrell 2026). The blended routing signal at β=0.4 achieves +11.72% quality improvement with Gini improving 11.8% — a better tradeoff point than any fixed β in the 0.1-0.3 range.

The most striking result is the recovery of under-discovered specialists. Specialist experts in Math and Reasoning received far fewer routing calls than optimal under static routing — the best available expert for a given token type was consistently bypassed in favor of General. The power metric health signal progressively corrects this as $P_i(t)$ builds evidence that specialists outperform General experts for math tokens 83% of the time. The real-time health signal corrects this without any retraining.

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### Contact & Collaboration


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Primary target collaborator: Azalia Mirhoseini (Stanford / Ricursive, AlphaChip, Intelligence Per Watt). MoE routing connects directly to her work on hardware-aware neural architecture and the Intelligence Per Watt framework. Proposed experiment: implement cross-expert P_i(t) on Mixtral-8x7B inference and measure quality + utilization improvement. Framework and patent available on request.